

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization with Power BI and Tableau

Practical – X

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Class: TYIT	Batch: 1
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Data Preparation and Transformation using Tableau Prep

a) Connect and load data into Tableau Prep

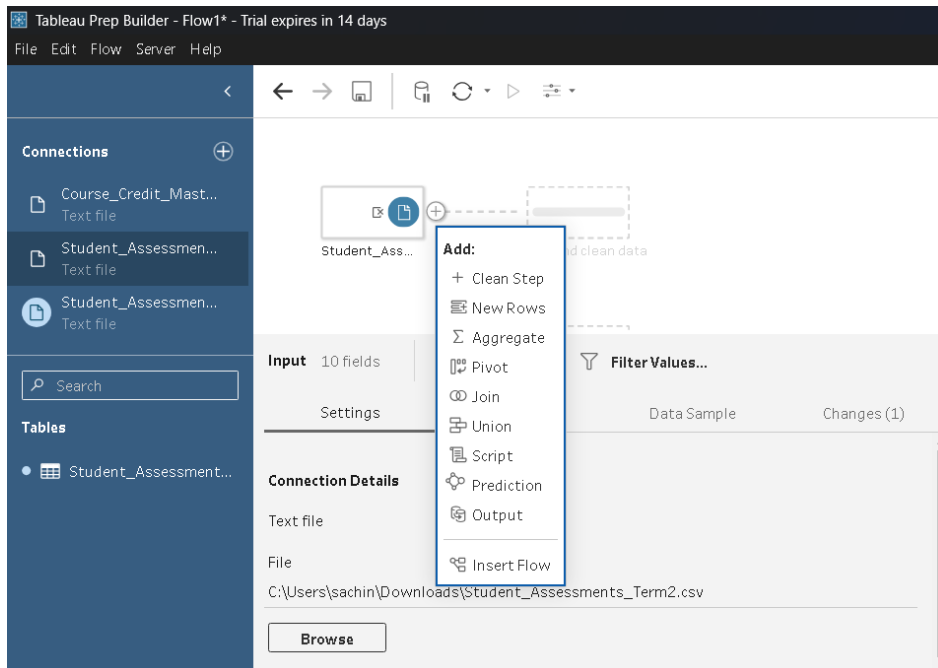
Step 1: Select connect to data and click on the text file to upload all three file and the data will be loaded and shown.

The screenshot displays the Tableau Prep interface. On the left, the 'Connections' pane lists three text files: 'Course_Credit_Mast...', 'Student_Assessmen...', and 'Student_Assessmen...'. The 'Tables' pane shows 'Student_Assessment...'. The main workspace shows three data sources connected to a 'View and clean data' step. The bottom pane shows the 'Input' section with '10 fields' and a 'Data Preview' tab. The preview shows a table with columns: '#', 'Source Row Number', 'Std_ID', 'Full_Name', 'Dept', and 'Exam_Type'. The data is for 'Student_Assessments_Term22'.

#	Source Row Number	Std_ID	Full_Name	Dept	Exam_Type
1	1	S101	Aarav Sharma	CS	Midterm
2	2	S101	Aarav Sharma	Computer Science	Final
3	3	S102	Priya Verma	ECE	Midterm
4	4	S102	Priya Verma	Electronics	Final
5	5	S103	Rohan Gupta	CSE	Midterm
6	6	S103	Rohan Gupta	CS	Final
7	7	S104	Sneha Patel	IT	Midterm
8	8	S104	Sneha Patel	Information Tech	Final
9	9	S105	Aditya Rao	Mech Engg	Midterm
10	10	S105	Aditya Rao	Mechanical	Final
11	11	S106	Ananya Singh	CS	Midterm
12	12	S106	Ananya Singh	CSE	Final

b) Clean and profile data

Step 1: Click on the Clean steps.



Step 2: After clicking on it the data will have changes shown below.

The screenshot shows the Tableau Prep Builder interface after the 'Clean 1' step. The flow diagram shows 'Student_Ass...' connected to 'Clean 1'. The 'Clean 1' step has 9 fields and 18 rows. The data is displayed in a table with columns: Student_ID, Student_Name, Department, Assessment..., Exam_Date, and Subject_Code. The data is as follows:

Student_ID	Student_Name	Department	Assessment...	Exam_Date	Subject_Code
S101	Aarav Sharma	Comp Sci	Final	15-09-2025	CS101
S102	aditya Rao	Computer Science	Midterm	16-09-2025	CS201
S103	Ananya Singh	CS		20-11-2025	EC101
S104	Karan Mehta	CSE		2025.09.15	EC201
S105	Neha Joshi	ECE		2025.09.16	IT101
S106	priya VERMA	Electronics		2025.11.20	IT201
S107	Rohan Gupta	Electronics Engg		2025.11.21	ME101
S108	Sneha Patel	Info Tech		2025/09/15	
S109	Vikram Shah	Information Tech		2025/09/16	
		IT		2025/11/20	
		Mech Engg		2025/11/21	
		Mechanical		21-11-2025	

Step 3: After changes we need to see the Student_Name card and click on ... we will click on clean → Trim Spaces

Clean 1 9 fields 18 rows

Select All Filter Values... Identify Duplicate Rows Automatic Split

Changes (0)

Student_ID 9

Student... 9

Department 12

Assessment... 2

Exam_Date 12

Filter

Clean

Group Values

Split Values

Identify Duplicate Rows

View State

Detail

Summary

Rename Field

Duplicate Field

Keep Only Field

Create Calculated Field

Hide Field

Remove

Make Uppercase

Make Lowercase

Make Titlecase

Remove Letters

Remove Numbers

Remove Punctuation

Trim Spaces

Remove Extra Spaces

Remove All Spaces

Student_ID	Student_Name	Department	Assessment_Type	Exam_Date	Subject_Code	Marks_Obtained	Max_Marks
S101	Aarav Sharma	Computer Science	Midterm	2025/09/15	CS101	85	100
S101	Aarav Sharma	Comp Sci	Final	2025/11/20	CS101	92	100
S102	priya VERMA	Electronics	Midterm	15-09-2025	EC101	70	100
S102	priya VERMA	ECE	Final	20-11-2025	EC101	78	100

Step 4: Now you can see changes in data it removes and trims unwanted spaces

Clean 1 9 fields 18 rows

Select All Filter Values... Identify Duplicate Rows Rename Fields...

Changes (1)

Student_ID 9

Student_Name 9

Department 12

Assessment... 2

Exam_Date 12

Subject_Code 7

Student_ID	Student_Name	Department	Assessment_Type	Exam_Date	Subject_Code	Marks_Obtained	Max_Marks	Enrollment_Status
S101	Aarav Sharma	Computer Science	Midterm	2025/09/15	CS101	85	100	Active
S101	Aarav Sharma	Comp Sci	Final	2025/11/20	CS101	92	100	Active
S102	priya VERMA	Electronics	Midterm	15-09-2025	EC101	70	100	Active
S102	priya VERMA	ECE	Final	20-11-2025	EC101	78	100	Active

Step 5: Now again go on Student_Name card and click on ... we will click on clean → Make Uppercase.

Clean 1 9 fields 18 rows

Select All Filter Values... Identify Duplicate Rows Automatic Split

Changes (1)

Student_ID 9

Student... 9

Department 12

Assessment... 2

Exam_Date 12

Filter

Clean

Group Values

Split Values

Identify Duplicate Rows

View State

Detail

Summary

Rename Field

Duplicate Field

Keep Only Field

Create Calculated Field

Hide Field

Remove

Make Uppercase

Make Lowercase

Make Titlecase

Remove Letters

Remove Numbers

Remove Punctuation

Trim Spaces

Remove Extra Spaces

Remove All Spaces

Student_ID	Student_Name	Department	Assessment_Type	Exam_Date	Subject_Code	Marks_Obtained	Max_Marks
S101	Aarav Sharma	Computer Science	Midterm	2025/09/15	CS101	85	100
S101	Aarav Sharma	Comp Sci	Final	2025/11/20	CS101	92	100
S102	priya VERMA	Electronics	Midterm	15-09-2025	EC101	70	100
S102	priya VERMA	ECE	Final	20-11-2025	EC101	78	100

Step 6: See changes in data

Student_ID 9	Student_Name 9	Departme... 12	Assessment_... 2	Exam_Date 12
S101	AARAV SHARMA	Comp Sci	Final	15-09-2025
S102	ADITYA RAO	Computer Science	Midterm	16-09-2025
S103	ANANYA SINGH	CS		20-11-2025
S104	KARAN MEHTA	CSE		2025.09.15
S105	NEHA JOSHI	ECE		2025.09.16
S106	PRIYA VERMA	Electronics		2025.11.20
S107	ROHAN GUPTA	Electronics Engg		2025.11.21
S108	SNEHA PATEL	Info Tech		2025/09/15
S109	VIKRAM SHAH	Information Tech		2025/09/16
		IT		2025/11/20
		Mech Engg		2025/11/21
		Mechanical		21-11-2025

Step 7: Now click on **Exam_Date** shown in **Exam_Date** change it into **Date**

Exam_Date 12

15-09-2025

16-09-2025

20-11-2025

2025.09.15

2025.09.16

2025.11.20

2025.11.21

2025/09/15

2025/09/16

2025/11/20

2025/11/21

21-11-2025

Assessment_... 2

Final

Midterm

Data Type

Number (decimal) #

Number (whole) #

Date & Time 📅

Date 📅

✓ String Abc

Data Role

✓ None

Email

URL

Geographic ▶

Step 8: Now see the changes

Exam_Date 4
15/09/2025
16/09/2025
20/11/2025
21/11/2025

Step 9: Now open **Student_Assessments_Term2** and Clean Step

Add:

- + Clean Step
- New Rows
- Aggregate
- Pivot
- Join
- Union
- Script
- Prediction
- Output
- Insert Flow

Student_ID	Student_Name
S101	AARAV SHARMA
S102	ADITYA RAO
S103	ANANYA SINGH
S104	KARAN MEHTA
S105	NEHA JOSHI

Step 10: Now do same step as click on three dot of Full_Name and click on Trim Spaces

Clean 2 9 fields 18 rows

Select All Filter Values... Identify Duplicate Rows Automatic Split

Changes (0)

Std_ID	Full_Name	Dept	Exam_Type
S101	Aarav Sharma	CS	Midterm
S102	Aarav Sharma	Computer Science	Final
S103	Aarav Sharma	Computer Science	Final
S104	Aarav Sharma	Computer Science	Final
S105	Aarav Sharma	Computer Science	Final
S106	Aarav Sharma	Computer Science	Final
S107	Aarav Sharma	Computer Science	Final
S108	Aarav Sharma	Computer Science	Final
S109	Aarav Sharma	Computer Science	Final

Filter Clean Group Values Split Values Identify Duplicate Rows View State Detail Summary Rename Field Duplicate Field Keep Only Field Create Calculated Field Hide Field Remove

Make Uppercase Make Lowercase Make Titlecase Remove Letters Remove Numbers Remove Punctuation Trim Spaces Remove Extra Spaces Remove All Spaces

Step 11: See changes in Full_Name.

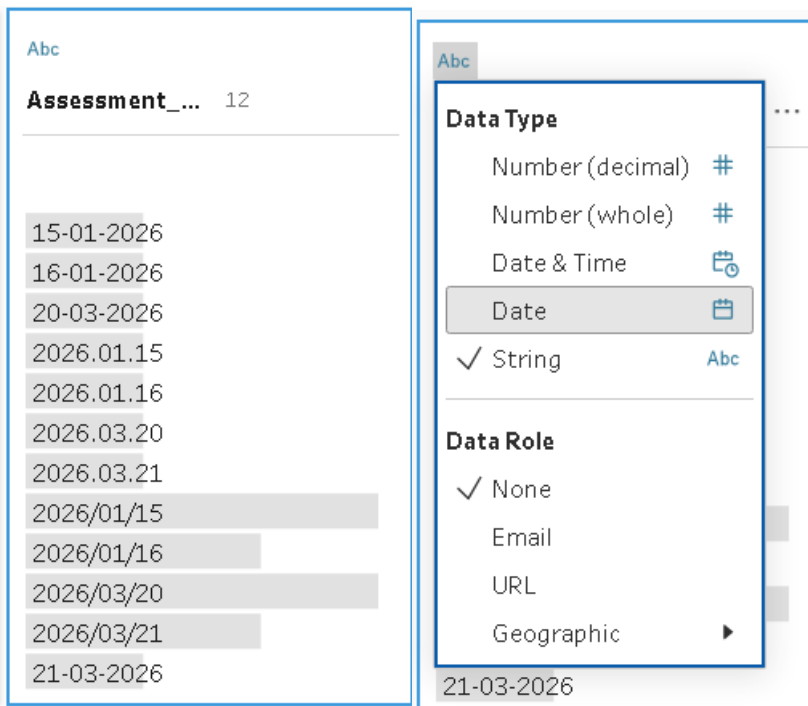
Clean 2 9 fields 18 rows

Select All Filter Values... Identify Duplicate Rows Automatic Split

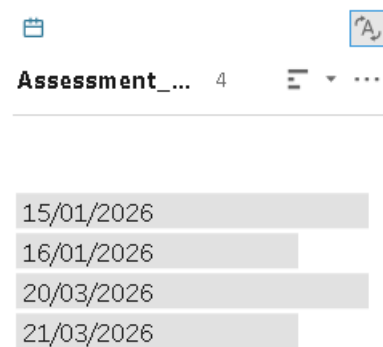
Changes (0)

Std_ID	Full_Name
S101	Aarav Sharma
S102	Aarav Sharma
S103	Aarav Sharma
S104	Aarav Sharma
S105	Aarav Sharma
S106	Aarav Sharma
S107	Aarav Sharma
S108	Aarav Sharma
S109	Aarav Sharma

Step 12: On Assessment_Date click on abc and change the data type to Date

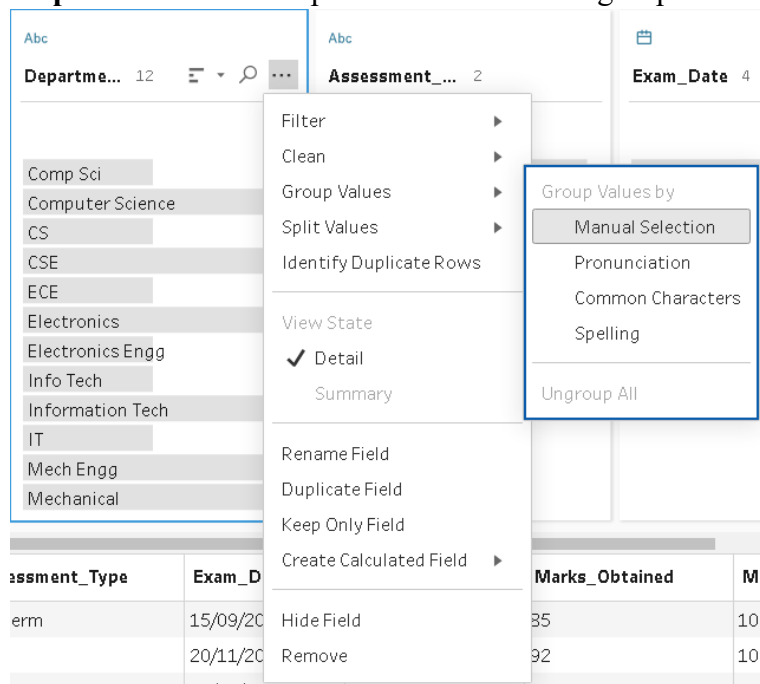


Step 13: See changes in Assessment_Date



c) Group and standardize values

Step 1: Select ... of Department and click on group values → Manual Selection



Step 2: Select manually all details you need in computer science and Electronics

Group Values by Manual Selection

Done

Department 10

Computer Science

CS

ECE

Electronics

Electronics Engg

Info Tech

Information Tech

IT

Mech Engg

Mechanical

Computer Science 3 members

☒ Comp Sci

☒ Computer Science

☒ CSE

☐ CS

☐ ECE

☐ Electronics

☐ Electronics Engg

☐ Info Tech

☐ Information Tech

☐ IT

☐ Mech Engg

☐ Mechanical

Information Tech
2 (11%) rows

Group Values by Manual Selection

Done

Department 8

Computer Science

CS

Electronics

Info Tech

Information Tech

IT

Mech Engg

Mechanical

Electronics 3 members

☒ ECE

☒ Electronics

☒ Electronics Engg

☐ Computer Science

☐ CS

☐ CS

☐ CS

☐ Mech Engg

☐ Mechanical

CS
1 (6%) row

Step 3: Now select other as shown below in Information Tech and for mechanical as well

Group Values by Manual Selection

Done

Deapar... 5

Computer Science

CS

Electronics

Information Tech

Mechanical

Information Tech 3 members

☒ Info Tech

☒ Information Tech

☒ IT

☐ Computer Science

☐ CS

☐ Electronics

☐ Mechanical

Step 4: Now see the changes in department showing 4 columns named Computer Science, Electronics, Information tech, Mechanical.

Abc

Department 4

Computer Science

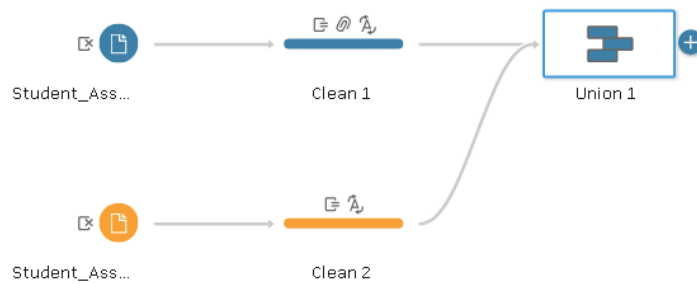
Electronics

Information Tech

Mechanical

d) Merge mismatched fields

Step 1: Create a union by moving Clean 2 to Clean 1



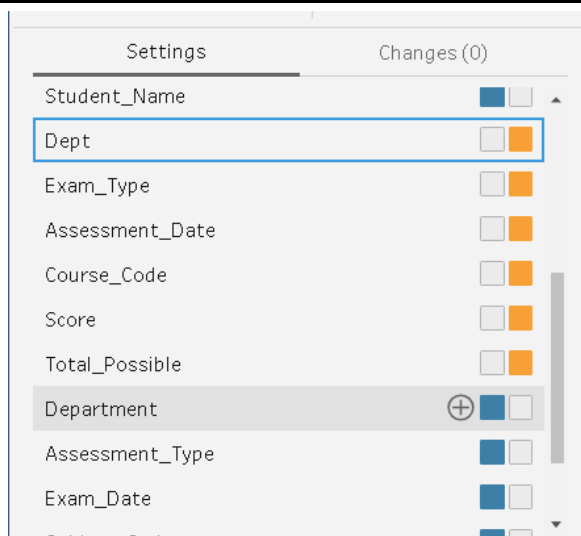
Step 2: Below you saw the mismatched fields you need to drag and add the same like Student_ID to Std_ID, Dept to Department

The screenshot shows the 'Union 1' settings panel. At the top, it indicates '18 fields 36 rows'. Below this, there are tabs for 'Settings' and 'Changes (0)'. The 'Inputs' section shows two inputs: 'Clean 1' (blue square) and 'Clean 2' (orange square). The 'Resulting Fields' section shows '16 Mismatching fields from 18 resulting fields.' The 'Mismatched Fields' section lists three fields: 'Full_Name', 'Student_Name', and 'Std_ID'. Each field has two checkboxes: a blue one and an orange one. The 'Std_ID' field has a dropdown arrow next to the orange checkbox.

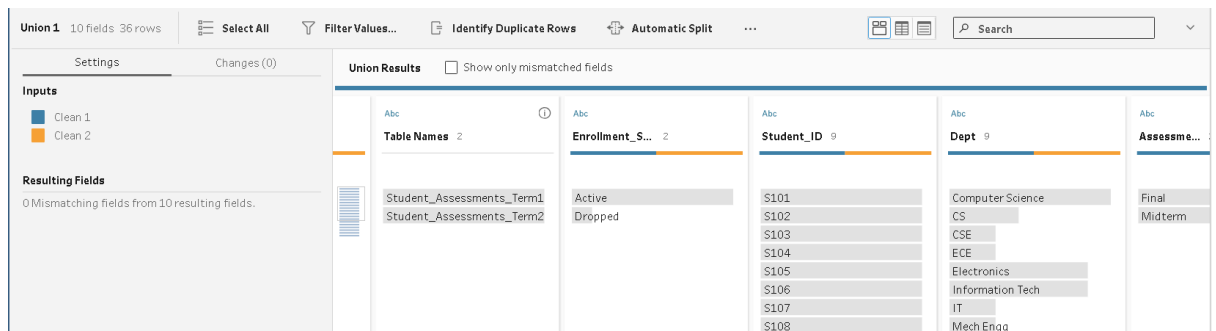
Step 3: You will saw the merged values of both Student_ID and Std_ID do the same for others as well

The screenshot shows a table with a header row and several data rows. The header row has a blue bar and an orange bar. The data rows show the following values: \$101, \$102, \$103, \$104, \$105, \$106, \$107, and \$108. The table is titled 'Student_ID' and has a search bar.

Student_ID
\$101
\$102
\$103
\$104
\$105
\$106
\$107
\$108

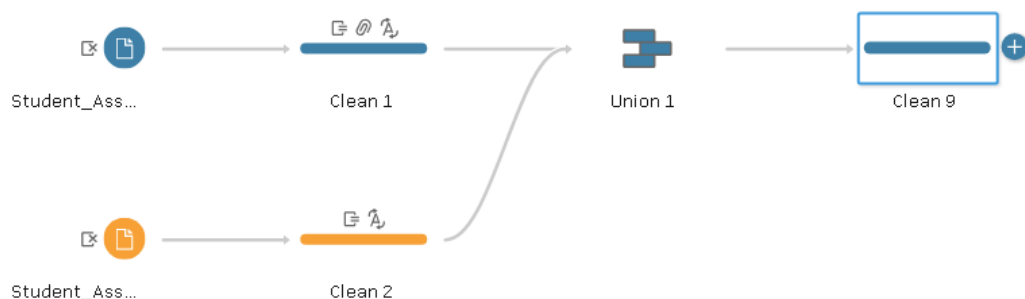


Step 4: Now click and do for all other mismatched files and you will see all done as shown below

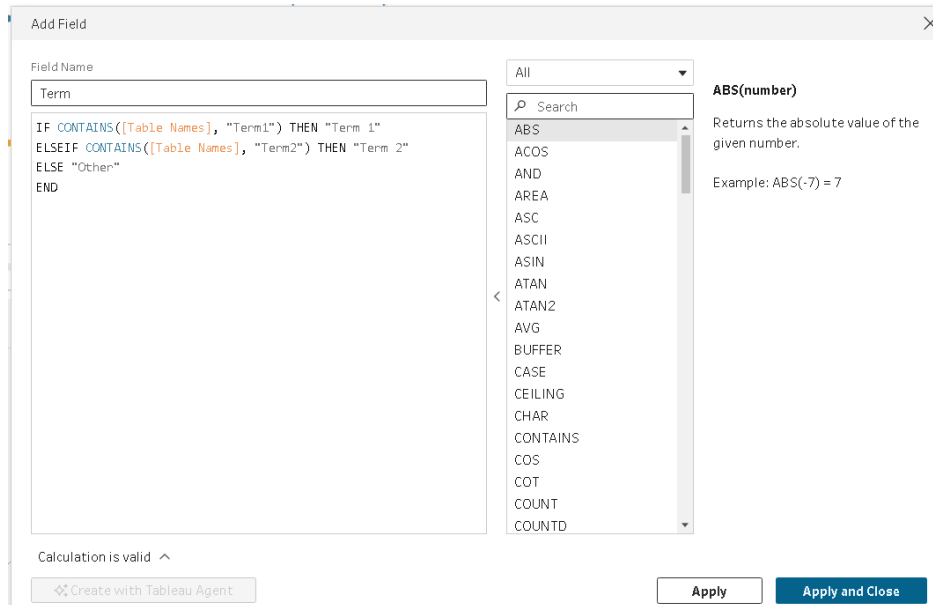


e) Perform aggregations in Prep

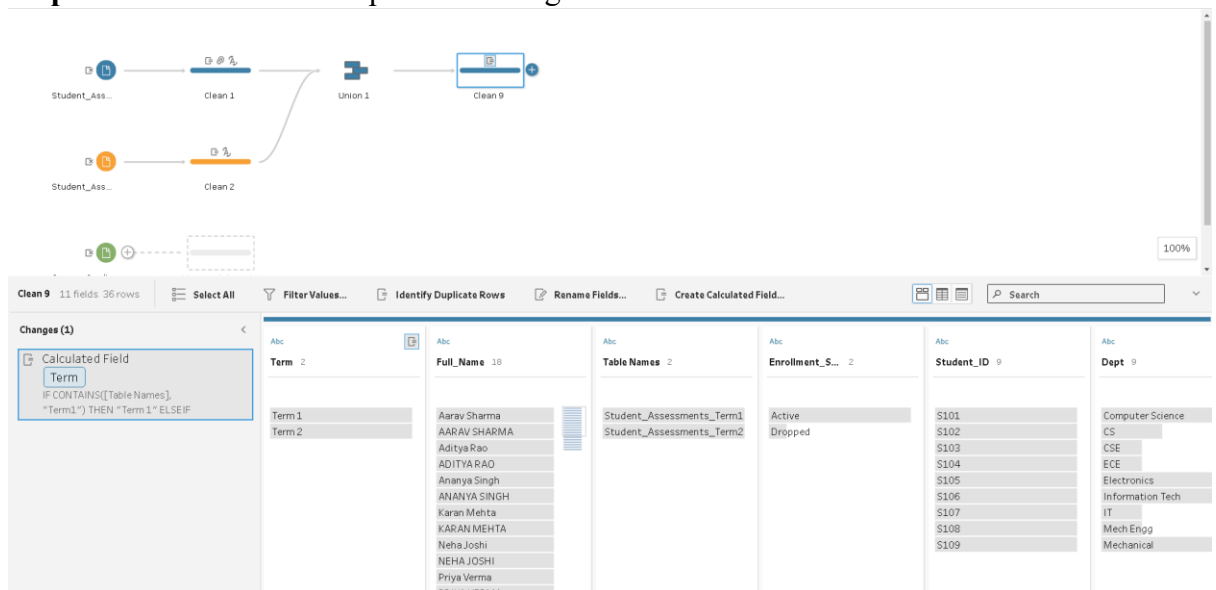
Step 1: Add clean step for Union 1.



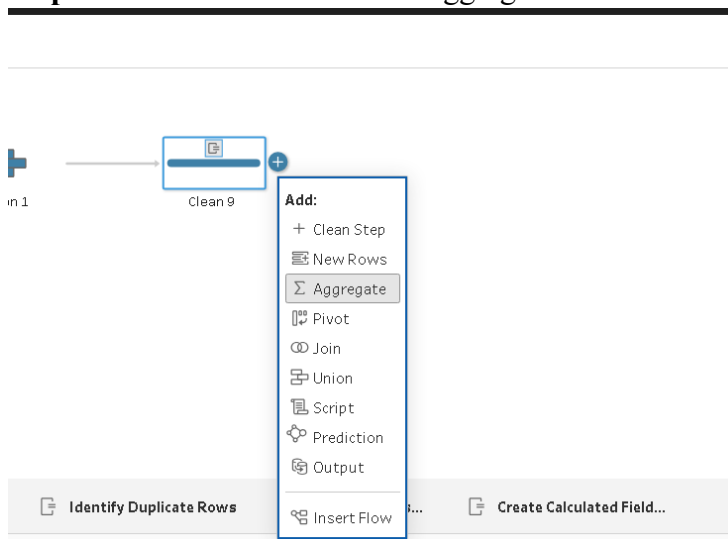
Step 2: Above the table you will see Create Calculated field and fill the details shown below as Field Name and Formula then Apply and Close.

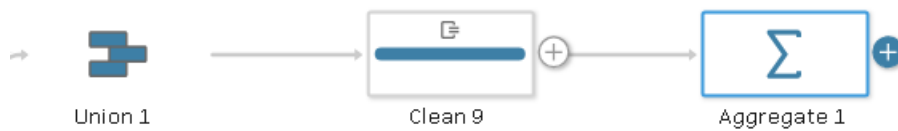


Step 3: You will see the output after changes shown below



Step 4: Select Clean 9 and add Aggregate .





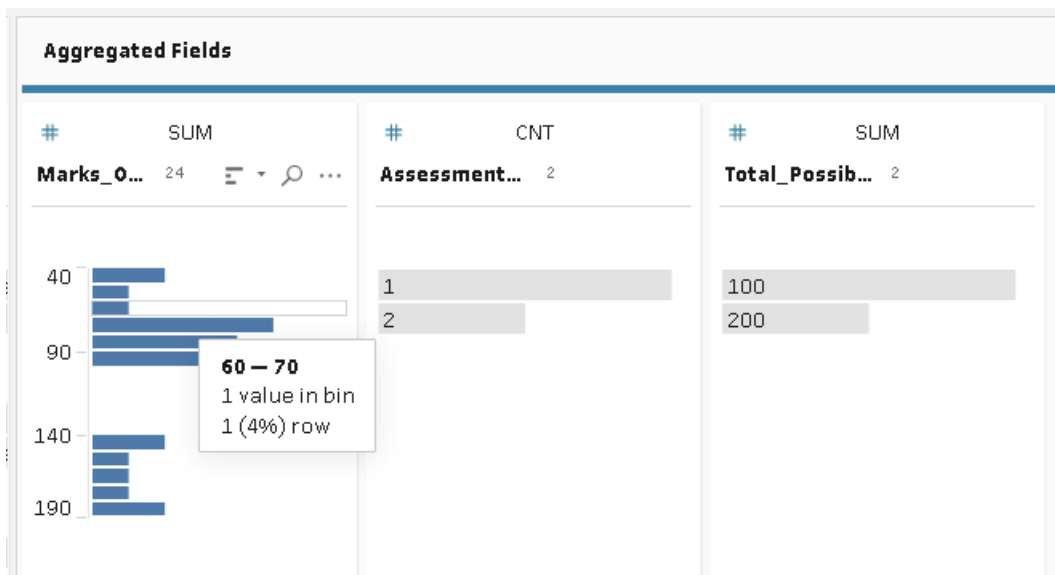
The screenshot shows the data tool interface with the following components:

- Workflow:** Student_Ass... -> Clean 1 -> Union 1 -> Clean 9 -> Aggregate 1.
- Aggregate 1 Configuration Panel:**
 - Settings:** Filter Values...
 - Additional Fields:** Drag fields to aggregate or group them. Includes a search bar and buttons for Add All and Remove All.
 - Grouped Fields:** Drop fields here to group them.
 - Aggregated Fields:** Drop fields here to aggregate them.

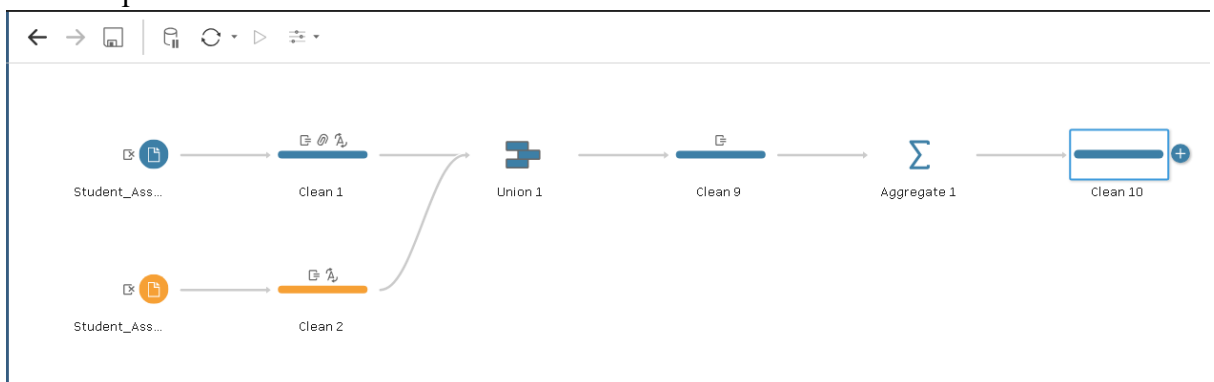
Step 5: Now add Term, Full Name, Student_ID, Dept

Grouped Fields			
Abc	GROUP	Abc	GROUP
Term 2	Full_Name 18	Student_ID 9	Dept 9
Term1	Aarav Sharma	S101	Computer Science
Term 2	AARAV SHARMA	S102	CS
	Aditya Rao	S103	CSE
	ADITYA RAO	S104	ECE
	Ananya Singh	S105	Electronics
	ANANYA SINGH	S106	Information Tech
	Karan Mehta	S107	IT
	KARAN MEHTA	S108	Mech Engg
	Neha Joshi	S109	Mechanical
	NEHA JOSHI		
	Priya Verma		
	PRIYA VERMA		

Step 6: Add the other three in Aggregate Fields as well.



Step 7: Now in Aggregate 1 we need to click on Clean Step and the Clean 10 will be created open it



Step 8: Below show the final output.

#	Abc	Abc	Abc	Abc	#
Marks_Obtai...	Term	Full_Name	Student_ID	Dept	Assessment_...
24	2	18	9	9	2
40	Term 1	Aarav Sharma	S101	Computer Science	1
90	Term 2	AARAV SHARMA	S102	CS	2
140		Aditya Rao	S103	CSE	
190		ADITYA RAO	S104	ECE	
		Ananya Singh	S105	Electronics	
		ANANYA SINGH	S106	Information Tech	
		Karan Mehta	S107	IT	
		KARAN MEHTA	S108	Mech Engg	
		Neha Joshi	S109	Mechanical	
		NEHA JOSHI			
		Priya Verma			
		PRIYA VERMA			

f) Apply filters and transformations

Step 1: Click on the three dot of Full_Name then Filter→Selected values.

Aggregate 1

- Filter
 - Selected Values
 - Wildcard Match
 - Null Values
- Clean
- Group Values
- Split Values
- Identify Duplicate Rows
- View State
 - Detail
 - Summary
- Rename Field
- Duplicate Field
- Keep Only Field
- Create Calculated Field
- Hide Field
- Remove

Full_Name	Student_ID	Dept
Aarav Sharma	S101	Computer Science
AARAV SHARMA	S102	CS
Aditya Rao	S103	CSE
ADITYA RAO	S104	ECE
Ananya Singh	S105	Electronics
ANANYA SINGH	S106	Information Tech
Karan Mehta	S107	IT
KARAN MEHTA	S108	Mech Engg
Neha Joshi	S109	Mechanical
NEHA JOSHI		
Priya Verma		
PRIYA VERMA		

Step 2: Filter the inactive/Dropped Student record and done

Filter: Selected Values Done

Full_Name 6 Keep Only 6 Exclude

Search

- ☒ Aarav Sharma
- ☒ AARAV SHARMA
- ☒ Aditya Rao
- ☒ ANANYA SINGH
- ☒ NEHA JOSHI
- ☒ PRIYA VERMA
- ☐ ADITYA RAO
- ☐ Ananya Singh
- ☐ Karan Mehta
- ☐ KARAN MEHTA

Select All Clear All

Step 3: Write the field name and formula shown below

Add Field

Field Name

Term_Percentage.

ROUND([Total_Marks_Obtained] / [Total_Max_Marks]) * 100, 2)

Calculation is valid

Create with Tableau Agent

Apply Apply and Close

ABS(number)

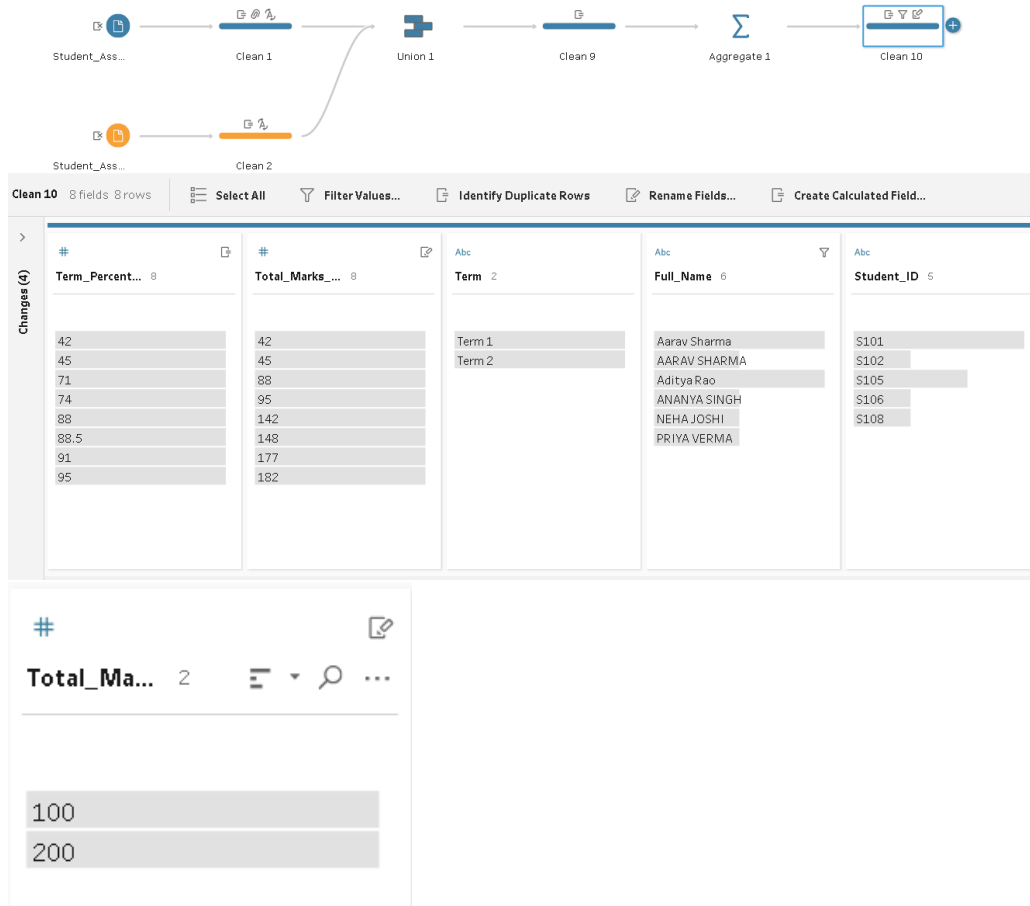
Returns the absolute value of the given number.

Example: ABS(-7) = 7

Search

- ABS
- ACOS
- AND
- AREA
- ASC
- ASCI
- ASIN
- ATAN
- ATAN2
- AVG
- BUFFER
- CASE
- CEILING

Step 4: See the changes below



Step 5: Add Letter_Grade and its formula

Add Field

Field Name: Letter_Grade

IF [Term_Percentage] >= 85 THEN "A (Distinction)"
 ELSEIF [Term_Percentage] >= 75 THEN "B (First Class)"
 ELSEIF [Term_Percentage] >= 60 THEN "C (Second Class)"
 ELSEIF [Term_Percentage] >= 40 THEN "D (Pass Class)"
 ELSE "F (Fail / Re-appear)"
 END

Calculation is valid

Create with Tableau Agent

Apply Apply and Close

ABS(number)
 Returns the absolute value of the given number.
 Example: ABS(-7) = 7

Step 6: Add Academic_Standing and formula as shown below

Add Field

Field Name

Academic_Standing

IF [Term_Percentage] >= 60 THEN "Satisfactory"

ELSE "Academic Probation (<60%)"

END

Calculation is valid

Create with Tableau Agent

All

Search

ABS
ACOS
AND
AREA
ASC
ASCII
ASIN
ATAN
ATAN2
AVG
BUFFER
CASE
CEILING
CHAR
CONTAINS
COS
COT
COUNT
COUNTD

ABS(number)

Returns the absolute value of the given number.

Example: ABS(-7) = 7

Apply

Apply and Close

Step 7: Now you will see the changes after adding the above formulas

Clean 10 10 fields 8 rows

Select All

Filter Values...

Identify Duplicate Rows

Automatic Split

Custom

Letter_Grade 3	Term_Percent... 8	Academic_Sta... 2	Total_Marks_... 8
A (Distinction)	42	Academic Probation (<60%)	42
C (Second Class)	45	Satisfactory	45
D (Pass Class)	71		88
	74		95
	88		142
	88.5		148
	91		177
	95		182

g) Create and automate data preparation flows

Step 1: Click on the clean 10 and click on + option and click on output

Clean 10

+

Add:

+ Clean Step

New Rows

Aggregate

Pivot

Join

Union

Script

Prediction

Output

Insert Flow

Calculated Field...

Term 2

Full_Name 6


```
C:\Windows\System32>dir "C:\Users\sachin\OneDrive\Dokumen\My Tableau Prep Repository\*.tfl" /s
Volume in drive C is Windows-SSD
Volume Serial Number is D4E0-1C17

Directory of C:\Users\sachin\OneDrive\Dokumen\My Tableau Prep Repository\Flows

04-09-2026  12:43                28,734 Practical 10.tfl
04-09-2026  12:43                28,734 Student_Performance_Prep_Flow.tfl
                2 File(s)              57,468 bytes

Total Files Listed:
                2 File(s)              57,468 bytes
                0 Dir(s)  189,394,382,848 bytes free
```

```
C:\Windows\System32>"C:\Program Files\Tableau\Tableau Prep Builder 2026.2\scripts\tableau-prep-cli.bat" -t "C:\Users\sachin\OneDrive\Dokumen\My Tableau Prep Repository\Flows\Practical 10.tfl"
JAVA_HOME is set to : C:\Program Files\Tableau\Tableau Prep Builder 2026.2\scripts\..\Plugins\jre temporarily
The connection file is in v1 format.
Preparing to run the flow : C:\Users\sachin\OneDrive\Dokumen\My Tableau Prep Repository\Flows\Practical 10.tfl
Loading the flow.
Creating temp directory at C:\Users\sachin\AppData\Local\Temp\prep-cli-15998479297145271073
Product license verified.
Loaded the flow.
checking the flow document for errors.
Flow Document has no errors.
Preparing to execute the flow.
Flow Execution Status: Running
Flow Execution Status: Finished
Finished running the flow successfully.
C:\Windows\System32>
```

Step 5: Now open the file in tableau desktop and add the column and rows as shown below and you will see the final output.

